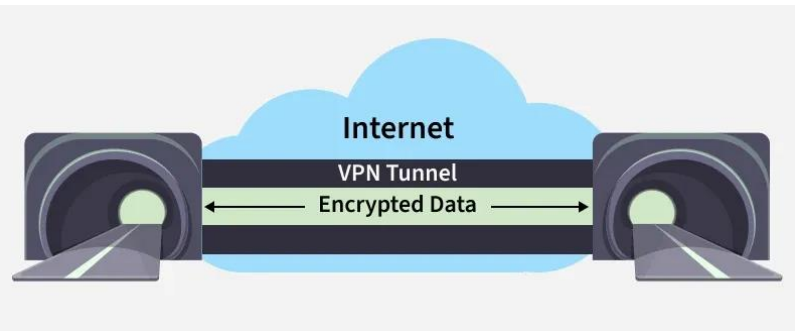


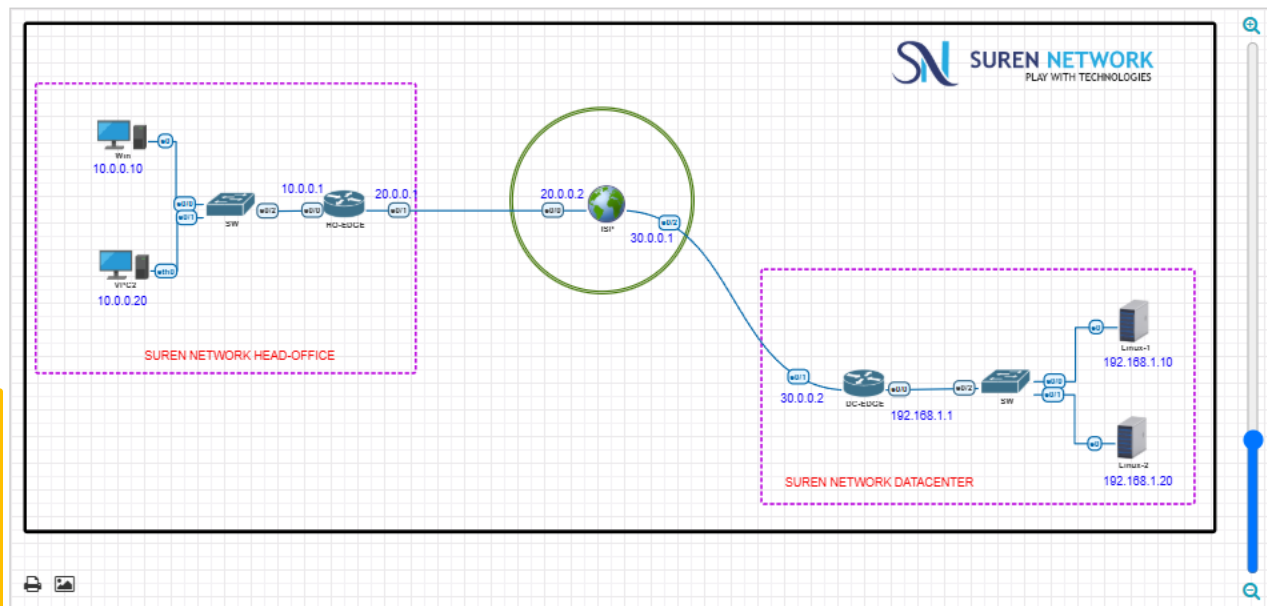
Search Labs	
☐	..
☐	1- LAB ACL.unl 30 Aug 2026 05:59
☐	4-IPV6.unl 19 Aug 2026 22:28
☐	2-LAB NAT.unl 29 Dec 2025 04:05
☒	3-LAB VPN.unl

A Virtual Private Network (VPN) is a security technology that creates an encrypted tunnel between your device and a VPN server over the internet, so your traffic travels privately and your real IP address is hidden.

Ex:-



Lab Preview

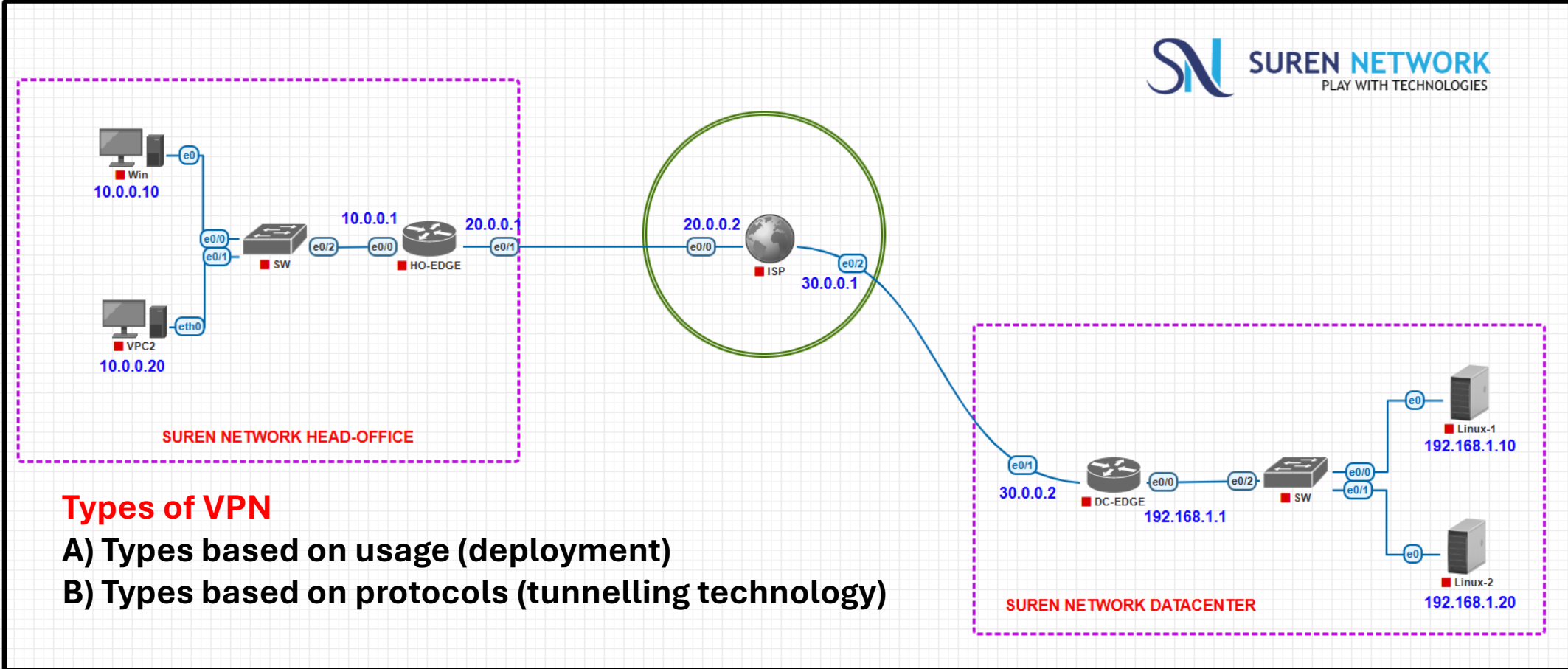


Open Edit Delete

- Lab Name : 3-LAB VPN
- Author : 1
- Version : 1
- Lab ID : 40c84e2f-7387-4424-a5be-f458520b5174
- Description :

CCNA CORE Subject
Title: Virtual Private Network

By Meganadhan M A



```

setting up system clock [utc] Fri Sep 18 13:43:52 UTC 2026 ..... [ OK ]
initializing random number generator ..... [ OK ]
setting up firewall ..... [ OK ]
startup klogd ..... [ OK ]
startup syslogd ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
udhcpd (v1.19.3) started
Sending discover...
Sending discover...
Sending discover...
No lease, forking to background
starting DHCP for Ethernet interface eth0 ..... [ OK ]
startup dropbear ..... [ OK ]
Starting thttpd HTTP server:
startup thttpd ..... [ OK ]
startup inetd ..... [ OK ]
startup gpm ..... [ OK ]
startup crond ..... [ OK ]

```

```
ttylinux ver 14.0 [hawberk]
i486 class Linux kernel 2.6.34.6 (/dev/tty1)
The initial "root" and "user" password is "password".
ttylinux_host login: root
Password:
```

Step 1

```

startup syslogd ..... [ OK ]
bringing up loopback interface lo ..... [ OK ]
udhcpd (v1.19.3) started
Sending discover...
Sending discover...
Sending discover...
No lease, forking to background
starting DHCP for Ethernet interface eth0 ..... [ OK ]
startup dropbear ..... [ OK ]
Starting tthttpd HTTP server:
startup tthttpd ..... [ OK ]
startup inetd ..... [ OK ]
startup gpm ..... [ OK ]
startup crond ..... [ OK ]

```

```
ttylinux ver 14.0 [hawberk]
i486 class Linux kernel 2.6.34.6 (/dev/tty1)
The initial "root" and "user" password is "password".
ttylinux_host login: root
Password:
```

Chop wood, carry water.

```
# vi /etc/sysconfig/network-scripts/ifcfg-eth0
```

Step 2

PNETLab | Topology Terminal GRE.cdr nat

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWZlZjpbeyJuYW11IjoiSE8tRURHRSlzInVybClk

Router Interface IP configured details

HO-EDGE x Win x DC-EDGE x VPC2 x Linux-2 x Linux-1 x

```
HO-EDGE#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              10.0.0.1        YES TFTP    up          up
Ethernet0/1              20.0.0.1        YES TFTP    up          up
Ethernet0/2              unassigned      YES TFTP    down        down
Ethernet0/3              unassigned      YES TFTP    down        down
HO-EDGE#
```

NAT commands using the default route IP

```
HO-EDGE#config t
Enter configuration commands, one per line.  End with CNTL/Z.
HO-EDGE(config)#
HO-EDGE(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
HO-EDGE(config)#
HO-EDGE(config)#
```

VPN Enable command is called **tunnel 1**

```
HO-EDGE(config)#int tunnel 1
HO-EDGE(config-if)#
*Sep 18 13:40:34.225: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down
HO-EDGE(config-if)#ip add 1.0.0.1 255.0.0.0
HO-EDGE(config-if)#tunnel source 20.0.0.1
HO-EDGE(config-if)#tunnel destination 30.0.0.2
HO-EDGE(config-if)#
*Sep 18 13:41:51.856: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
HO-EDGE(config-if)#ex
HO-EDGE(config)#
HO-EDGE(config)#ip route 192.168.1.0 255.255.255.0 tunnel 1
HO-EDGE(config)#
```

Tunnel cmd for VPN enabling

VPN Tunnel IP

Source IP

Destination IP

Pls, add a route to the server's private IP

```
H0-EDGE(config)#do sh int tunnel 1
Tunnell is up, line protocol is up
  Hardware is Tunnel
  Internet address is 1.0.0.1/8
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 20.0.0.1, destination 30.0.0.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:06:39
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
H0-EDGE(config)#
```

PNETLab | Topology x Terminal x GRE.cdr x nat

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW11lloiSE8tRURHRSlInVyl

HO-EDGE x Win x DC-EDGE x VPC2 x Linux-2 x Linux-1 x

```
DC-EDGE#sh ip int br
Interface                IP-Address      OK? Method Status Protocol
Ethernet0/0              192.168.1.1     YES TFTP   up      up
Ethernet0/1              30.0.0.2        YES TFTP   up      up
Ethernet0/2              unassigned      YES TFTP   down    down
Ethernet0/3              unassigned      YES TFTP   down    down
DC-EDGE#
DC-EDGE#config t
Enter configuration commands, one per line.  End with CNTL/Z.
DC-EDGE(config)#
DC-EDGE(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.1
DC-EDGE(config)#
```

```
DC-EDGE(config)#
DC-EDGE(config)#int tunnel 1
DC-EDGE(config-if)#
*Sep 18 13:44:48.300: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down
DC-EDGE(config-if)#ip add 1.0.0.2 255.0.0.0
DC-EDGE(config-if)#tunnel source 30.0.0.2
DC-EDGE(config-if)#tunnel destination 20.0.0.1
DC-EDGE(config-if)#e
*Sep 18 13:45:36.922: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
DC-EDGE(config-if)#ex
DC-EDGE(config)#
DC-EDGE(config)#ip route 10.0.0.0 255.0.0.0 tunnel 1
DC-EDGE(config)#sh int tunnel 1
      ^
% Invalid input detected at '^' marker.
```

Types of VPN

A) Types based on usage (deployment)

VPN Based on Usage



Remote
Access VPN



Site-to-Site
VPN



Mobile VPN



MPLS VPN

B) Types based on protocols (tunnelling technology)

VPN Based on Protocols



PPTP



L2TP/IPsec

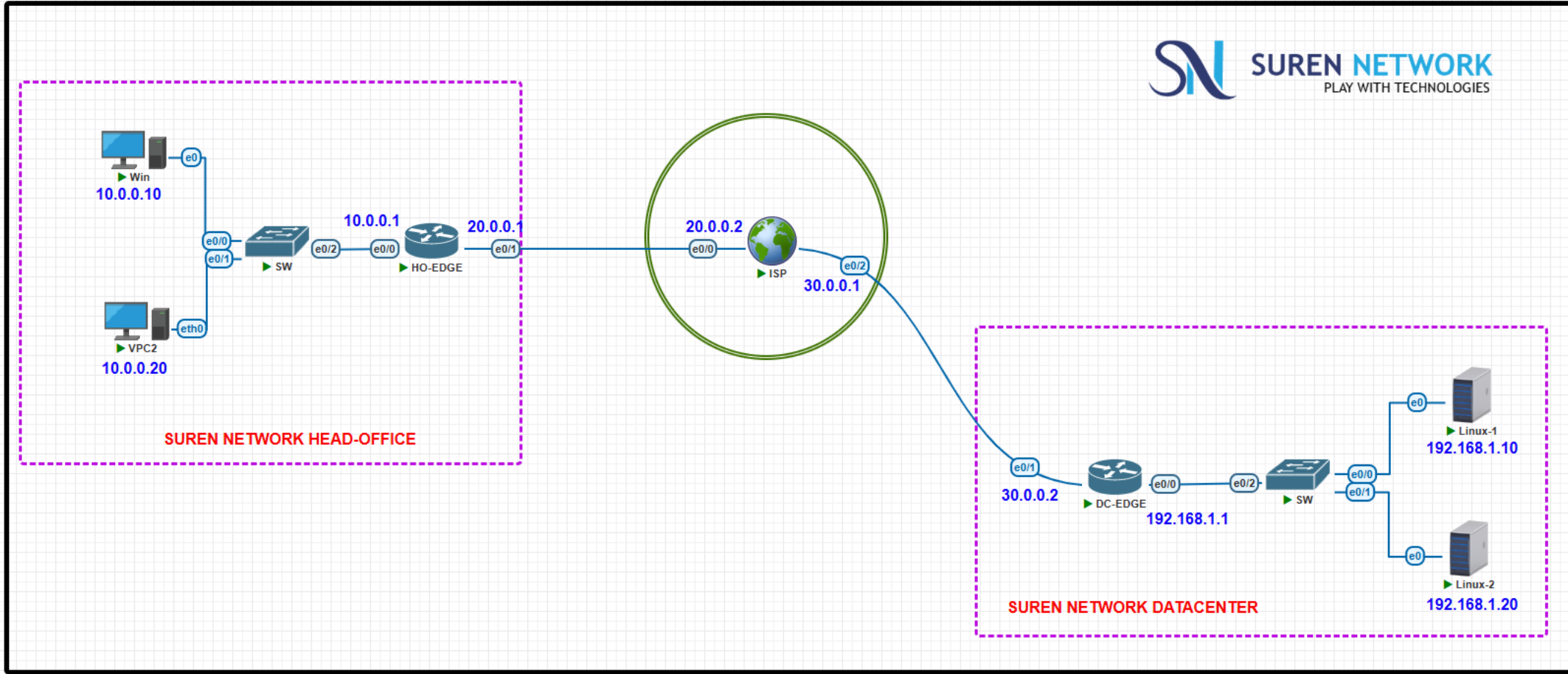


OpenVPN



IKEv2/IPsec

```
DC-EDGE(config)#do sh int tunnel 1
Tunnell is up, line protocol is up
Hardware is Tunnel
Internet address is 1.0.0.2/8
MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel linestate evaluation up
Tunnel source 30.0.0.2, destination 20.0.0.1
Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
Tunnel TTL 255, Fast tunneling enabled
Tunnel transport MTU 1476 bytes
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)
Last input never, output never, output hang never
Last clearing of "show interface" counters 00:02:11
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
DC-EDGE(config)#
*Sep 18 13:48:28.672: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Disco
DC-EDGE(config)#
```



To check the reachability from the client PC to the servers

```
HO-EDGE x Win x DC-EDGE x VPC2 x Linux-2 x Linux-1 x
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>cd\
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=3ms TTL=62
Reply from 192.168.1.10: bytes=32 time=2ms TTL=62
Reply from 192.168.1.10: bytes=32 time=1ms TTL=62
Reply from 192.168.1.10: bytes=32 time=1ms TTL=62

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=2ms TTL=62
Reply from 192.168.1.20: bytes=32 time=2ms TTL=62
Reply from 192.168.1.20: bytes=32 time=1ms TTL=62
Reply from 192.168.1.20: bytes=32 time=3ms TTL=62

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>ping 10.0.0.20

Pinging 10.0.0.20 with 32 bytes of data:

Reply from 10.0.0.20: bytes=32 time=1ms TTL=64
Reply from 10.0.0.20: bytes=32 time=1ms TTL=64
Reply from 10.0.0.20: bytes=32 time=1ms TTL=64
Reply from 10.0.0.20: bytes=32 time<1ms TTL=64

Ping statistics for 10.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ping Tests: Show a successful ICMP ping from Win (10.0.0.10) or VPC2 (10.0.0.20) to Linux-1 (192.168.1.10).

```
HO-EDGE x Win x DC-EDGE x VPC2 x Linux-2 x Linux-1 x
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC2>
VPC2> ip 10.0.0.20 255.0.0.0 10.0.0.1
Checking for duplicate address...
VPC2 : 10.0.0.20 255.0.0.0 gateway 10.0.0.1

VPC2> sh ip

NAME       : VPC2[1]
IP/MASK     : 10.0.0.20/8
GATEWAY     : 10.0.0.1
DNS         : 
MAC         : 00:50:79:66:68:71
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

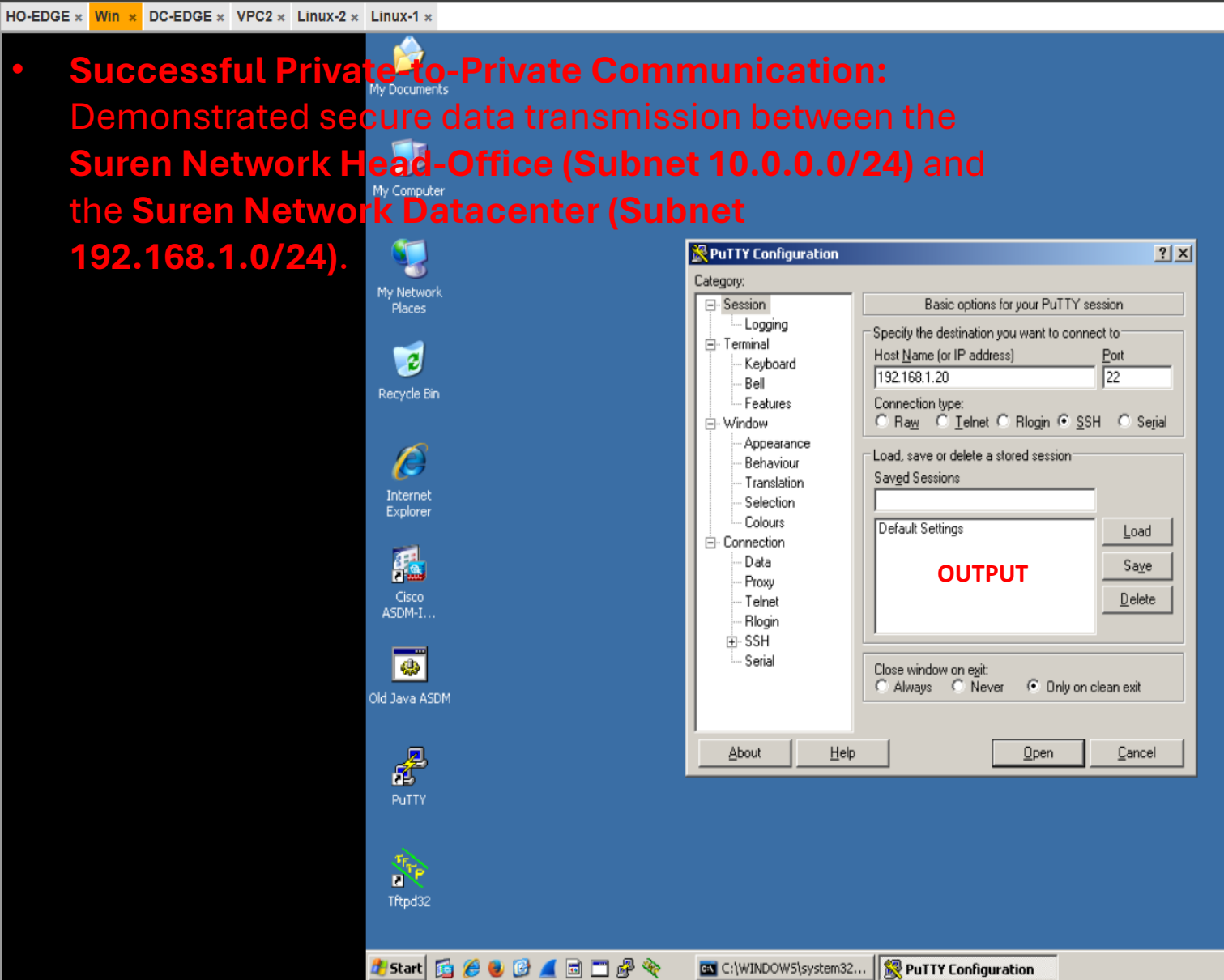
VPC2> ping 192.168.1.20

84 bytes from 192.168.1.20 icmp_seq=1 ttl=62 time=1.696 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=62 time=1.944 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=62 time=2.141 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=62 time=2.123 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=62 time=2.261 ms

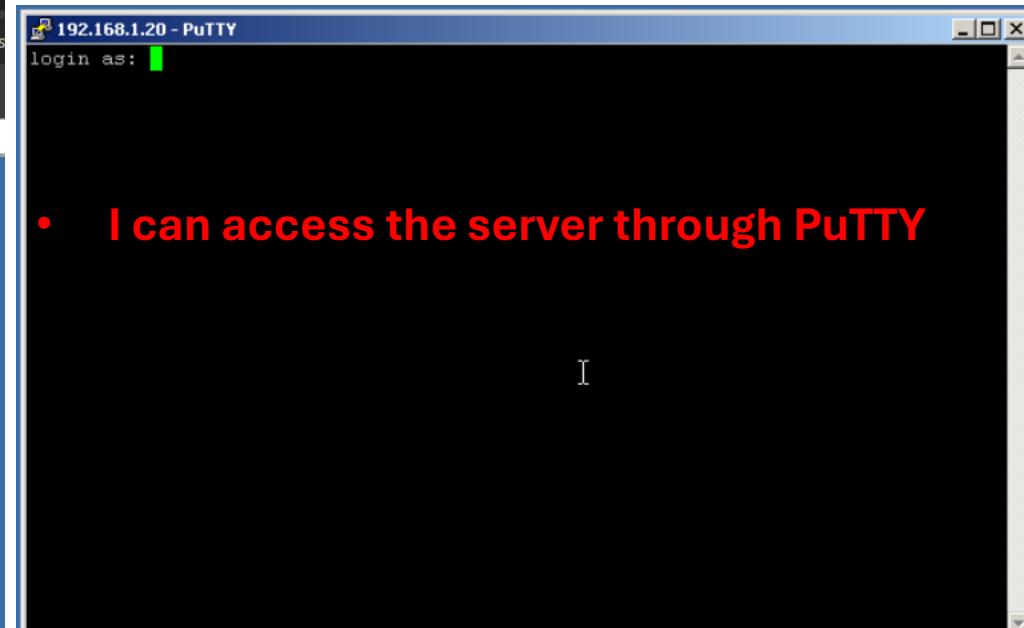
VPC2> ping 192.168.1.10

192.168.1.10 icmp_seq=1 timeout
192.168.1.10 icmp_seq=2 timeout
192.168.1.10 icmp_seq=3 timeout
192.168.1.10 icmp_seq=4 timeout
192.168.1.10 icmp_seq=5 timeout

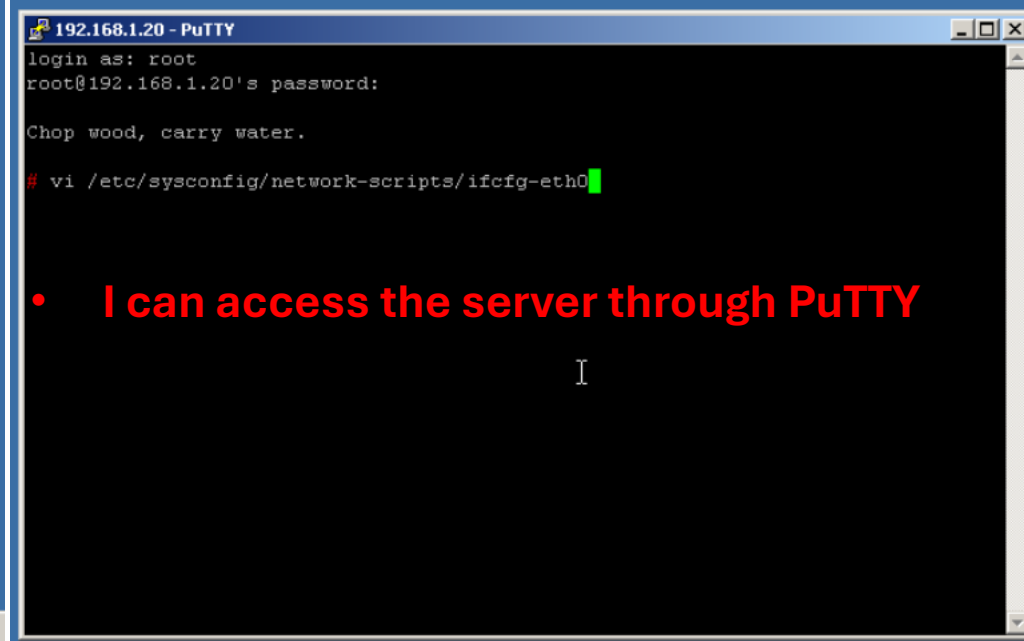
VPC2>
```



- **Successful Private-to-Private Communication:** Demonstrated secure data transmission between the **Suren Network Head-Office (Subnet 10.0.0.0/24)** and the **Suren Network Datacenter (Subnet 192.168.1.0/24)**.



- **I can access the server through PuTTY**



- **I can access the server through PuTTY**

The screenshot shows a PNETLab terminal window with the following tabs: PNETLab | Topology, Terminal, and GRE.cdr. The address bar shows the URL: labs.surenetwork.com/store/public/admin/labs/terminal?terminals=eyJ0Y... The terminal window has a tab labeled VPC2. The output of the 'sh ip' command is shown, with a red arrow pointing to the 'OUTPUT' label. The output is as follows:

```
VPC2> sh ip
NAME       : VPC2[1]
IP/MASK    : 10.0.0.20/8
GATEWAY    : 10.0.0.1
DNS        :
MAC        : 00:50:79:66:68:71
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
```

The output of the 'ping' command is shown for three destinations:

```
VPC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=2.778 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=3.062 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=3.565 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=2.292 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=2.193 ms

VPC2> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=62 time=1.895 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=62 time=2.732 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=62 time=2.743 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=62 time=2.375 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=62 time=2.685 ms

VPC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=254 time=1.615 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=254 time=1.938 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=254 time=1.445 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=254 time=1.544 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=254 time=1.475 ms

VPC2> ping 30.0.0.2
84 bytes from 30.0.0.2 icmp_seq=1 ttl=254 time=1.844 ms
84 bytes from 30.0.0.2 icmp_seq=2 ttl=254 time=1.557 ms
84 bytes from 30.0.0.2 icmp_seq=3 ttl=254 time=1.388 ms
84 bytes from 30.0.0.2 icmp_seq=4 ttl=254 time=1.462 ms
84 bytes from 30.0.0.2 icmp_seq=5 ttl=254 time=2.640 ms

VPC2>
```

Types of VPN

A) Types based on usage (deployment)

- Remote Access VPN: It allows an individual user to securely connect to a private network over the internet, and it is widely used by employees working remotely.
- Site-to-Site VPN: It securely connects two or more separate networks, such as a head office and branch offices, so internal communication remains protected across locations.
- Mobile VPN: It is designed for mobile users and keeps the VPN session stable even when the device switches between Wi-Fi and cellular networks.
- MPLS VPN: It is a provider-managed enterprise WAN solution that offers scalable connectivity and traffic prioritization, but it typically does not provide end-to-end encryption by default.

B) Types based on protocols (tunnelling technology)

PPTP: It is an older protocol that can be fast, but it provides weak security, so it is mainly used only for legacy systems.

L2TP/IPsec: It combines L2TP tunnelling with IPsec encryption, which improves security, but it can add performance overhead.

OpenVPN: It is an open-source protocol that uses SSL/TLS for encryption, and it is widely adopted because it provides strong security and flexibility.

IKEv2/IPsec: It is a secure and fast protocol that works very well on mobile devices because it reconnects quickly when network conditions change.

VPN Protocols

- | | | |
|---------------|--------------|----------------|
| 1. OpenVPN | 2. WireGuard | 3. IKEv2/IPsec |
| 4. L2TP/IPsec | 5. PPTP | 6. SSTP |

INTERVLAN USING SVI

Step 1 : Create default route in head office router

```
HO-EDGE(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
```

Step 2 : Create gre vpn configuration in head office router

```
HO-EDGE(config)#interface tunnel 1
HO-EDGE(config-if)#ip add 1.0.0.1 255.0.0.0
HO-EDGE(config-if)#tunnel source 20.0.0.1
HO-EDGE(config-if)#tunnel destination 30.0.0.2
HO-EDGE(config-if)#exit
```

Step 3 : Create vpn route in head office router

```
HO-EDGE(config)#ip route 192.168.1.0 255.255.255.0 tunnel 1
```

Step 4 : Create default route in datacenter router

```
DC-EDGE(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.1
```

Step 5 : Create gre vpn configuration in datacenter router

```
DC-EDGE(config)#interface tunnel 1
DC-EDGE(config-if)#ip add 1.0.0.2 255.0.0.0
DC-EDGE(config-if)#tunnel source 30.0.0.2
DC-EDGE(config-if)#tunnel destination 20.0.0.1
DC-EDGE(config-if)#exit
```

Step 6 : Create vpn route in datacenter router

```
DC-EDGE(config)#ip route 10.0.0.0 255.0.0.0 tunnel 1
```

Step 7 : Check gre tunnel status

```
HO-EDGE # show interface tunnel 1
```

Step 8 : Check reachability from ho private network to datacenter private network